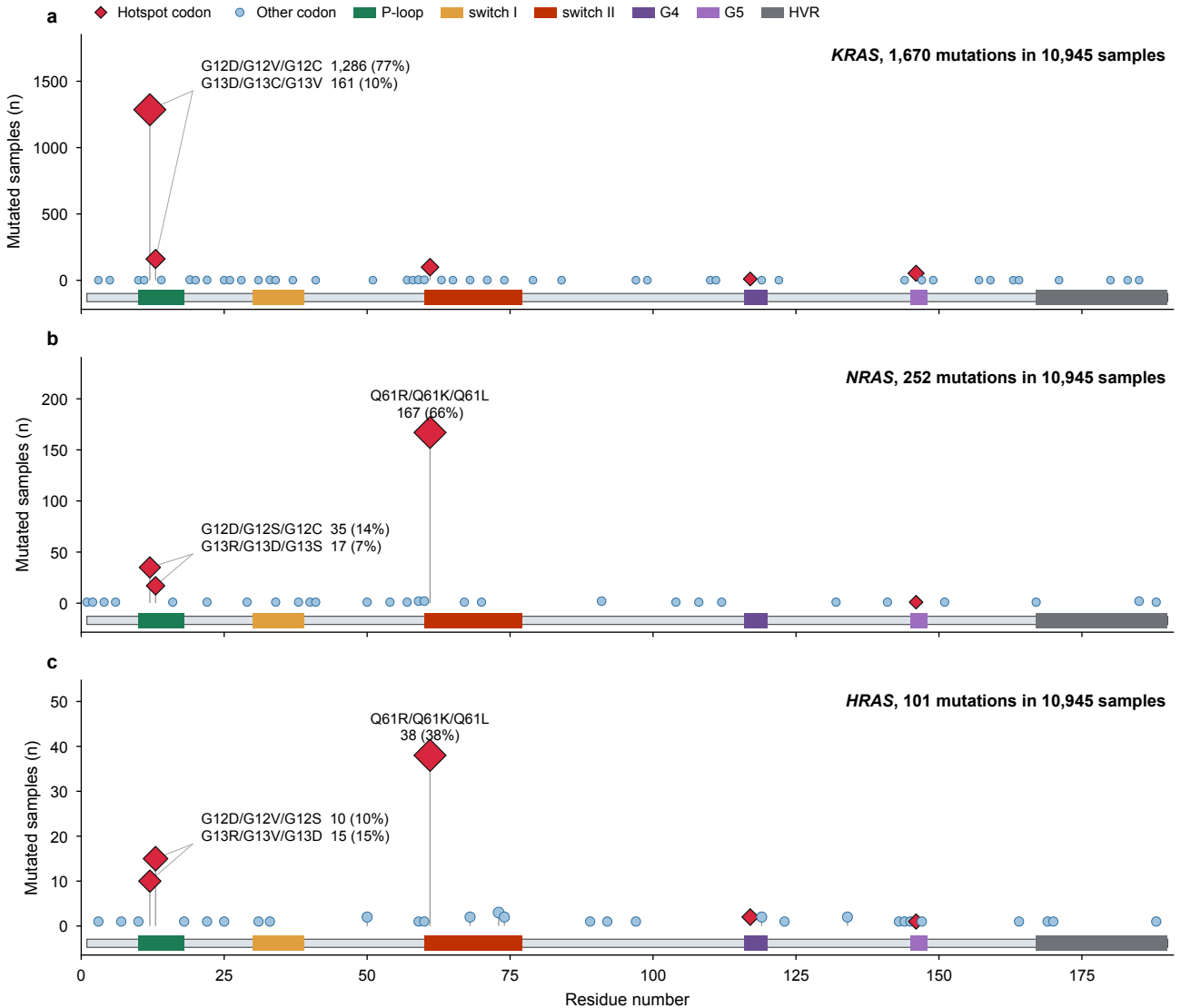

a
Hotspot codon
Other codon
P-loop
switch I
switch II
G4
G5
HVR
KRAS, 1,670 mutations in 10,945 samples
G12D/G12V/G12C 1,286 (77%)
G13D/G13C/G13V 161 (10%)
Mutated samples (n)
1500
1000
500
0
b
NRAS, 252 mutations in 10,945 samples
Q61R/Q61K/Q61L
167 (66%)
G12D/G12S/G12C 35 (14%)
G13R/G13D/G13S 17 (7%)
Mutated samples (n)
200
150
100
50
0
c
HRAS, 101 mutations in 10,945 samples
Q61R/Q61K/Q61L
38 (38%)
G12D/G12V/G12S 10 (10%)
G13R/G13V/G13D 15 (15%)
Mutated samples (n)
50
40
30
20
10
0
0
25
50
75
100
125
150
175
Residue number